

Hamiltonian
 $H = T + V, U_2(\delta t)$

Bounds

state-independent
 $c_{\text{bound}}, \|\mathcal{E}\|$

Exact

eigenphase of U_2
following ψ_0
 c_{exact}

Perturbative

$\langle \psi | \mathcal{E} | \psi \rangle$
needs a reference
state ψ

Step count

steps $\propto \sqrt{|c|}$
T gates = steps
 $\times T/\text{step}$

This paper

$\psi \in \{\psi_0, \text{free-fermion, Néel, Gutzwiller}\}$; family c_λ
estimate $c_{1/2}$, check $|r|/48$